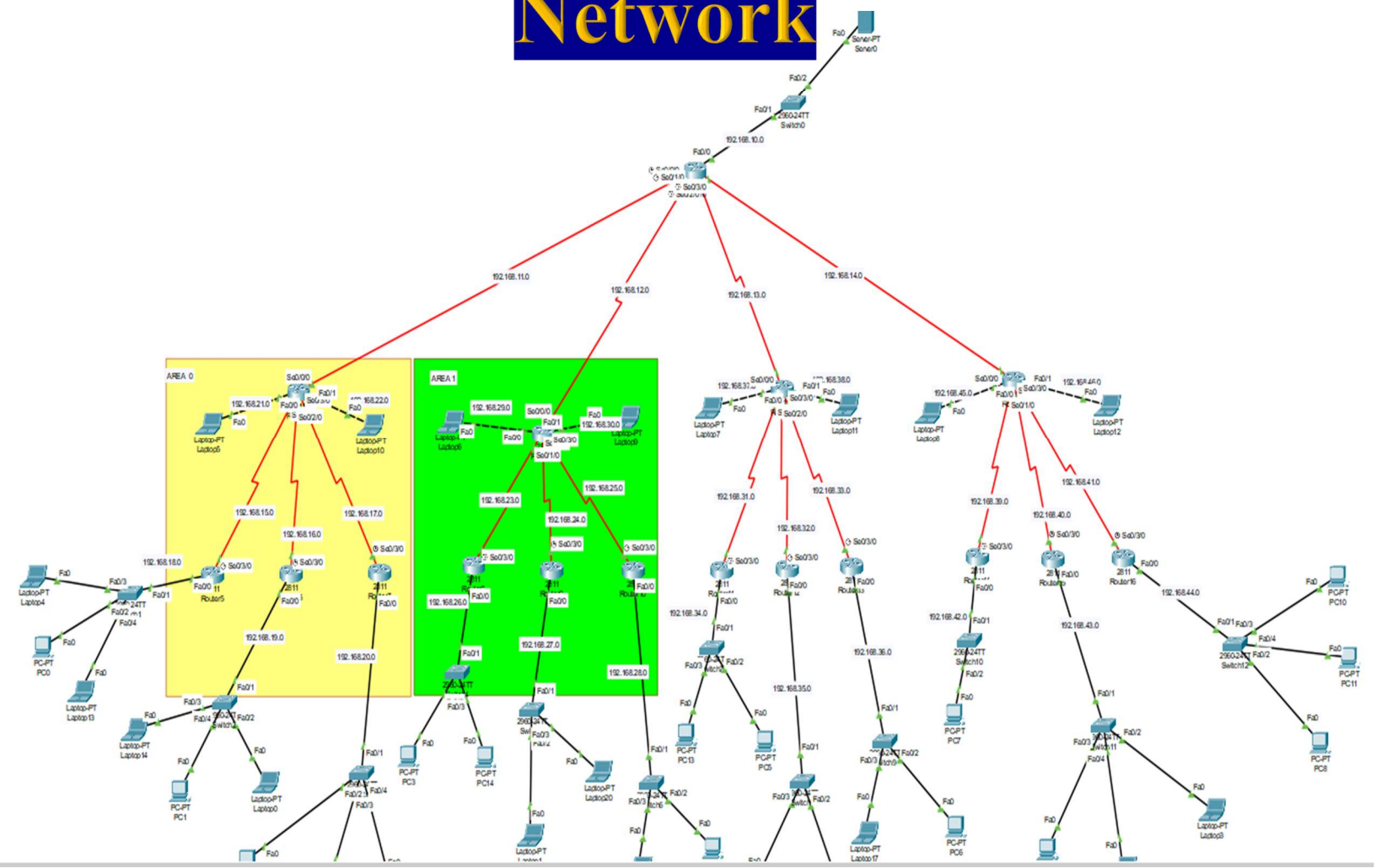
Multi-Store Retail Chain Network



Computer Networks Project - Fall 2024

RAJA SAIF ALI

I22-1353

Objective:

The primary objective of this project is to design and implement a robust and efficient network infrastructure for a multi-store retail chain. Each store operates as an independent subnet to ensure streamlined communication and management. The project aims to utilize DHCP for automated IP address allocation, and OSPF, RIP for optimized inter-store and data center connectivity. This setup ensures seamless network operations, scalability for future expansions, and enhanced security across the retail chain.

Technologies Used:

1. Cisco Packet Tracer (Instructor Version): Network simulation.
2. Networking Protocols:

   OSPF (Open Shortest Path First)

   RIP (Routing Information Protocol)
3. DHCP (Dynamic Host Configuration Protocol): Automated IP addressing.

Implementation Details:

Network Design and Approach

1. **Routing Protocols:**

   I have four districts and with each district 2 end devices are connected and with each district 3 shops are connected and with each shop 3 end devices are connected and then the four districts are connected to central data router and the central data router is connected to the server.

   OSPF (Area 0) and Area 1): First two districts (District 1 in Area 0) and (District 1 in area 1). Then redistribution is used in central router to connect them to other two areas where RIP protocol is used

   RIP: In other two districts (District 3 and District 4) RIP protocol is used and then again redistribution is used on central router to connect them to the districts where OSPF protocol is used

2. **Redistribution:**

   Redistribution is used on central data router as district routers communicate to each other through central data router.

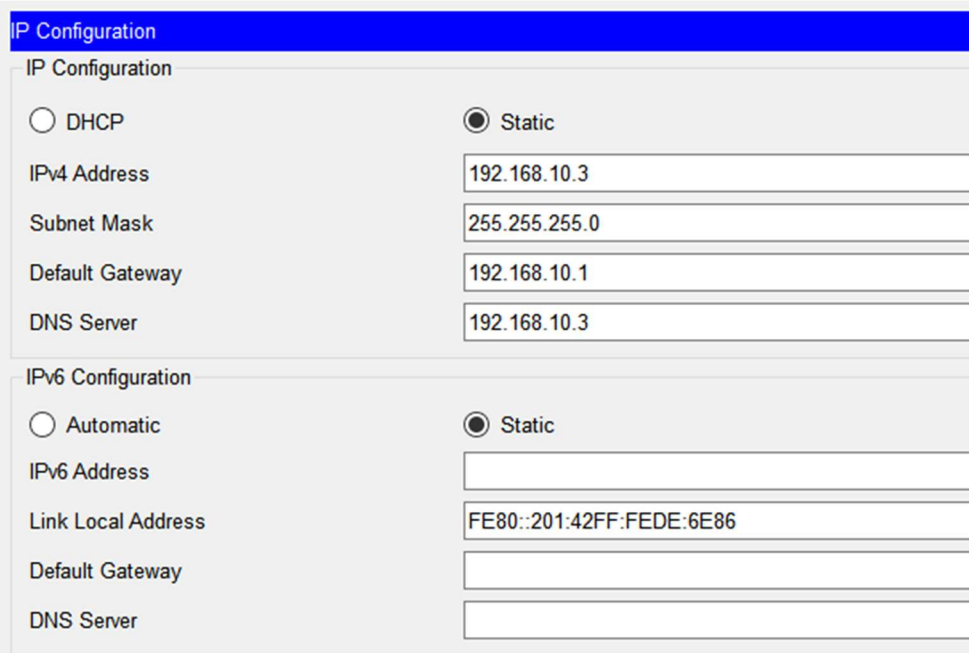
3. **DHCP Server:**

DHCP server is connected to central data router as all districts are connected to central data router. So it seamlessly assigns IP's to end devices.

Key Configuration Example

DHCP:

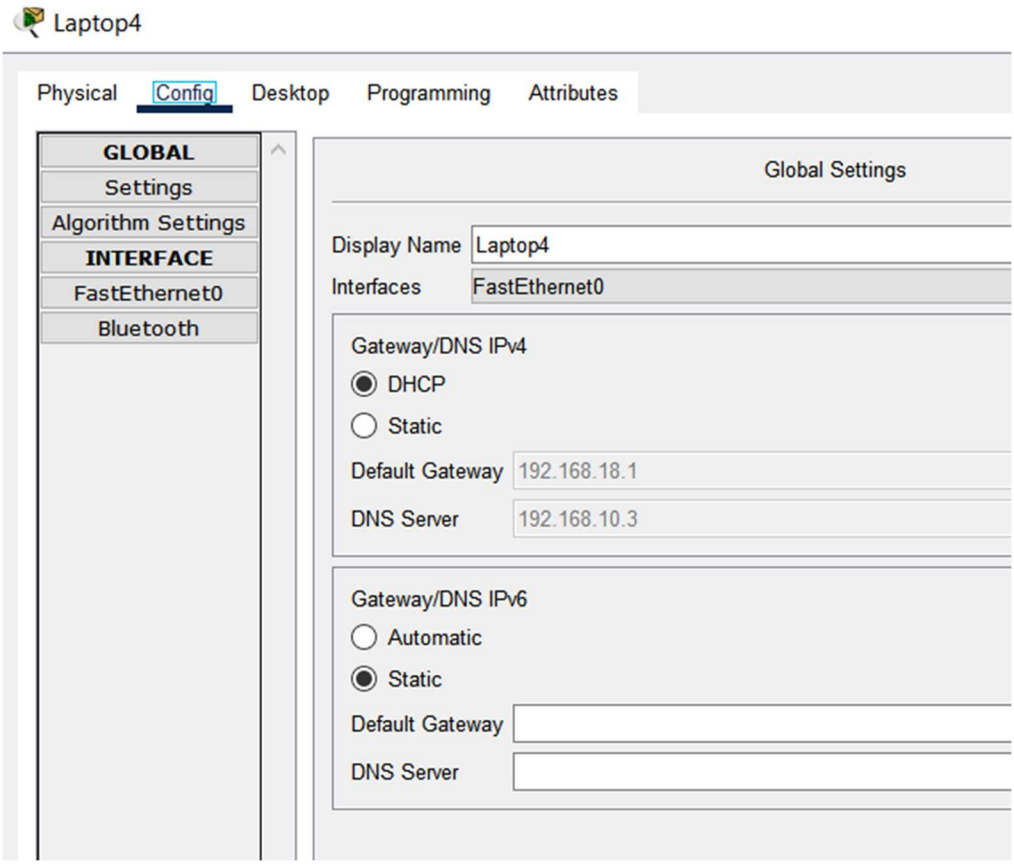
This is my server configuration



And these are some of the pools

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DR1	192.168.21.1	192.168.10.3	192.168.21.10	255.255.255.0	40	0.0.0.0	0.0.0.0
serverPool	192.168.10.1	192.168.10.3	192.168.10.10	255.255.255.0	40	0.0.0.0	0.0.0.0
DS3	192.168.20.1	192.168.10.3	192.168.20.10	255.255.255.0	40	0.0.0.0	0.0.0.0
DS2	192.168.19.1	192.168.10.3	192.168.19.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D2R2	192.168.30.1	192.168.10.3	192.168.30.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D2R1	192.168.29.1	192.168.10.3	192.168.29.10	255.255.255.0	40	0.0.0.0	0.0.0.0
DR2	192.168.22.1	192.168.10.3	192.168.22.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D2S1	192.168.26.1	192.168.10.3	192.168.26.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D2S2	192.168.27.1	192.168.10.3	192.168.27.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D2S3	192.168.28.1	192.168.10.3	192.168.28.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D3R1	192.168.37.1	192.168.10.3	192.168.37.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D3R2	192.168.38.1	192.168.10.3	192.168.38.10	255.255.255.0	40	0.0.0.0	0.0.0.0
D3S1	192.168.34.1	192.168.10.3	192.168.34.10	255.255.255.0	40	0.0.0.0	0.0.0.0

And this is an example of DHCP assigned IP to one of the end devices



Results and Testing

1. **Screenshots:**

District 1 Router OSPF

```
Router>sh ip pr

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.22.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.21.0 0.0.0.255 area 0
    192.168.22.0 0.0.0.255 area 0
    192.168.15.0 0.0.0.255 area 0
    192.168.16.0 0.0.0.255 area 0
    192.168.17.0 0.0.0.255 area 0
    192.168.11.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.14.1         110      00:00:10
    192.168.18.1         110      00:00:10
    192.168.19.1         110      00:00:09
    192.168.20.1         110      00:00:09
    192.168.22.1         110      00:00:09
  Distance: (default is 110)
 --More--
```

District 2 Router OSPF

```
Router>sh ip pr

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.30.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.12.0 0.0.0.255 area 1
    192.168.29.0 0.0.0.255 area 1
    192.168.30.0 0.0.0.255 area 1
    192.168.23.0 0.0.0.255 area 1
    192.168.24.0 0.0.0.255 area 1
    192.168.25.0 0.0.0.255 area 1
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.14.1         110      00:01:13
    192.168.26.1         110      00:01:12
    192.168.27.1         110      00:01:13
    192.168.28.1         110      00:01:12
    192.168.30.1         110      00:01:12
  Distance: (default is 110)
 --More--
```

District 3 Router RIP

```
Router>sh ip pr
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 23 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0       12 1
  FastEthernet0/1       12 1
  Serial0/0/0           12 1
  Serial0/1/0           12 1
  Serial0/2/0           12 1
  Serial0/3/0           12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
          192.168.13.0
          192.168.31.0
          192.168.32.0
          192.168.33.0
          192.168.37.0
 --More--
```

District 4 Router RIP

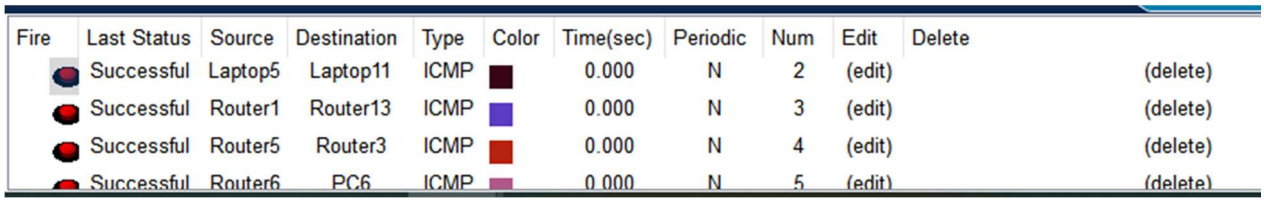
```
Router>sh ip pr
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 22 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface              Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0        12 1
  FastEthernet0/1        12 1
  Serial0/0/0            12 1
  Serial0/1/0            12 1
  Serial0/2/0            12 1
  Serial0/3/0            12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
          192.168.14.0
          192.168.39.0
          192.168.40.0
          192.168.41.0
          192.168.45.0
 --More--
```

Central Data Router (where redistribution is applied)

```
Router>sh ip pr
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 1 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, ospf 1
Default version control: send version 1, receive any version
  Interface              Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0        12 1
  Serial0/3/0            12 1
  Serial0/2/0            12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
          192.168.10.0
          192.168.13.0
          192.168.14.0
Passive Interface(s):
Routing Information Sources:
          Gateway         Distance      Last Update
          192.168.13.2         120      00:00:24
          192.168.14.2         120      00:00:23
 --More--
```

2. Testing:

Verified interconnectivity across subnets.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Laptop5	Laptop11	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Router1	Router13	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Router5	Router3	ICMP		0.000	N	4	(edit)	(delete)
	Successful	Router6	PC6	ICMP		0.000	N	5	(edit)	(delete)

Challenges and Learnings:

1. Challenges:

DHCP Configuration:

Configuring a centralized DHCP server to dynamically allocate IPs for multiple subnets.

Ensuring proper scope configuration and resolving IP conflicts.

OSPF and RIP Routing:

Setting up OSPF and RIP for efficient inter-store communication and data center connectivity.

Troubleshooting OSPF and RIP adjacency issues and understanding routing table discrepancies.

Device Configuration Errors:

Ensuring proper IP addressing, hostname assignment, and routing configurations for all devices.

Troubleshooting errors during device initialization or simulation.

2. Learnings:

Enhanced DHCP Skills:

Learned to configure and manage a centralized DHCP server for multi-subnet environments.

Proficiency in Routing Protocols:

Understood the principles of routing configuration and its impact on network efficiency.

Network Troubleshooting:

Developed skills in identifying and resolving network issues, such as misconfigurations or routing errors.

Simulation and Testing:

Learned the importance of testing network performance in simulated environments before deployment.

Hands-on Experience with Cisco Packet Tracer:

Improved proficiency in using Cisco Packet Tracer for designing and simulating complex network architectures.

Conclusion:

The Multi-Store Retail Chain Network project successfully demonstrates the design and implementation of a scalable, secure, and efficient network infrastructure. By leveraging DHCP, RIP and OSPF, the network ensures seamless communication between stores and the central data center while maintaining optimal performance and security.

The project highlighted the importance of meticulous planning and configuration to address challenges such as routing complexities, and IP management. Through this hands-on experience with Cisco Packet Tracer, valuable skills in network design, troubleshooting, and scalability planning were gained, paving the way for future real-world implementations.

This project serves as a robust foundation for building modern, interconnected networks that cater to the dynamic needs of multi-location businesses.
